Calibrated collision detection and selective memory for aerial robots among invisible obstacles

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I. Problem description and primary reference paper

Autonomous micro aerial vehicles (MAVs) use LiDAR to navigate cluttered indoor spaces. Their navigation depends on detecting obstacles, but two common cases leave gaps in perception. Glass walls, doors and windows, transparent curtains and thin wires can let laser beams pass through. The map then shows free space, and the planner directs the MAV into the obstacle. Obstacles can also enter from above or below the LiDAR's limited vertical field of view without being seen.

Perception alone cannot prevent collisions in these cases. The MAV needs to detect contact, react before it crashes, and remember the obstacle to avoid another hit.

Permanent memory creates a problem when a person moves away, an object is removed or a door opens. The map retains a phantom wall that can make the goal unreachable. False detections can block routes in the same way.

Our project addresses the suggested topic Robot Failure Detection and Recovery. We aim to detect MAV collisions at a known false-alarm rate and retain collision memories selectively: invisible static obstacles should remain in the map, while transient obstacles and false alarms should be forgotten.

Our primary reference is R. Wang *et al.*, "Air Bumper: A Collision Detection and Reaction Framework for Autonomous MAV Navigation," *IEEE ICRA 2024*, pp. 15735–15741 [1]. This original research paper was published in 2024 at ICRA, an approved venue, within the required period. Its caged MAV has a 3D LiDAR and uses the IMU as its only collision sensor. It detects collisions when any body-axis acceleration exceeds a fixed threshold ($a^* = 20\,\mathrm{m/s^2}$, sliding window $N = 10$), then estimates the collision direction and contact point on a spherical cage. It reacts immediately by setting a position away from the contact, blended with the gradient of the Euclidean distance transform (EDT), which gives the distance to known obstacles. The contact patch is added to the volumetric map as a "collision sensor" mark that LiDAR observations never clear.

The paper demonstrates recovery from glass doors in Gazebo and from glass and stick impacts on a real MAV. Across 10 real trials, detection takes approximately 0.18 s and recovery takes less than 1 s.

We chose Air Bumper because it is an early framework that combines collision detection, reaction and mapping in a standard LiDAR stack using only the IMU for collision sensing. It treats failure as a sensor by adding contact information to the map, which fits our project's focus on detection and recovery. The implementation is open source [2] and has been validated in simulation and on hardware. Its fixed threshold and permanent collision marks also give us specific design choices to investigate within a term project.

Since its preprint in July 2023, follow-up work has cited Air Bumper, including whisker-based tactile navigation in *Nature Communications* [3] and a survey of LiDAR-based aerial vehicles in *IEEE/ASME T-Mech* [4]. Semantic Scholar lists 7 citing papers in Nov. 2026. Related work [5] predicts planner failures from LiDAR geometry before they occur, while [6] adapts edge costs online using contact feedback. Both assume that LiDAR detects every obstacle, an assumption that fails with glass.

Indoor MAVs used for inspection, warehouse inventory and search and rescue encounter glass facades, windows and partitions, as well as people and moving objects. False reactions interrupt flight, and phantom walls block routes. We retain the paper's caged MAV platform and its detection, reaction and memory pipeline, focusing on how the system declares a collision and how long it remembers the contact.

II. Environment, benchmark, and dataset

No public benchmark contains invisible and transient obstacles for MAVs, so we will generate the scenario suite procedurally with fixed seeds, including a recreation of the paper's layout.

- S1: Static glass (60 worlds). Rooms and corridors with 1–4 glass walls, glass doors (recreating the three-door world of [1]) and windows at random positions and heights. Tests detection and memory for invisible static obstacles.
- S2: Transient (45 worlds). A glass door opens 20–120 s after the first contact; a visible panel on the shortest route is removed after contact; a pole hits the cage from below or above the LiDAR FOV (the paper's stick test). Tests phantom-wall failures.
- S3: False-alarm stress (30 worlds). Collision-free flights with wind gusts (WindEffects), aggressive stop-and-go and sharp turns, high-speed flight and doubled IMU noise. Measures false alarms per hour.

Each set is split by world into 50/20/30% train/calibration/test subsets. Gust and maneuver profiles absent from calibration form an out-of-distribution test. We use speeds $v_{\max} \in \{0.5, 1.0, 1.5\}$ m/s.

We plan to collect approximately 10 h of collision-free flight for calibration (Sec. IV), a separate collision-free test

TABLE I
SOFTWARE AND SIMULATION ENVIRONMENT (ALL FREE).

Component	Choice	Justification
Middleware	ROS 2 Jazzy [7] in Docker	A free LTS release that provides the same setup on every laptop
Autopilot	PX4 SITL [8] via uXRCE-DDS (`px4_msgs`), offboard position/velocity setpoints	Uses the same autopilot as [1]; reaction setpoints go directly to the low-level controller
Simulator	Gazebo Harmonic [9] (PX4 `gz` integration), headless batch runs	Uses the same simulator family as [1], provides ground truth at no cost and supports gusts through *WindEffects*
Vehicle	PX4 x500 quadrotor with a spherical collision cage (radius $l = 0.25$ m), 3D LiDAR configured like a Livox Mid-360 ($360^\circ \times 59^\circ$), IMU (250 Hz, noise + bias), cage contact sensor	Matches the paper's caged MAV; the contact sensor provides ground-truth labels and is not an input to the detector
Map & planner	Local voxel occupancy map (0.1 m) with EDT [10], A* path search, velocity-limited setpoint follower	A lightweight Python replacement for the paper's GPU mapper and GTO-MPC, with a cost and speed limit for each voxel
Invisible obstacles	Models with collision geometry but no visual geometry	The GPU LiDAR renders only visual geometry, so beams pass through while the physics engine registers contact, reproducing the glass setup in [1]
Moving / transient	Joint/trajectory-driven panels and poles, doors opening at random times, objects removed after contact, a "stick" hitting from outside the LiDAR FOV	Tests memories of obstacles that have disappeared and reproduces the paper's stick test

TABLE II
LIMITATIONS OF AIR BUMPER [1].

#	Limitation (evidence)	Why it matters
L1	Fixed, hand-tuned threshold on raw acceleration, $a^* = 20\,\mathrm{m/s^2}$ (Sec. IV-A, "experimentally determined"); no false-alarm or miss rates reported	Aggressive maneuvers and wind gusts also cause acceleration spikes, while soft or low-speed contacts stay below a^*. The false-alarm rate is unknown and varies with platform, speed and wind
L2	Per-axis test $\lvert a_x\rvert > a^*$ or $\lvert a_y\rvert > a^*$ (Sec. IV-A)	A diagonal hit splits the acceleration between axes: $25\,\mathrm{m/s^2}$ at 45° gives $17.7\,\mathrm{m/s^2}$ per axis, so the detector misses it
L3	Direction from raw acceleration, $\mathbf{a}_c = -\mathbf{a}$ (Eqs. 1–4)	Commanded acceleration biases the estimated direction, recovery setpoint and mapped patch, especially while braking or turning
L4	Permanent, top-priority marks: "the robot remembers all obstacles it ever collides with" (Sec. V-C)	Permanent marks retain static glass, but also leave phantom walls after people move, objects are removed or doors open. The map also preserves false detections from L1
L5	LiDAR evidence at the contact is ignored	A LiDAR return distinguishes visible obstacles, which are already mapped and may move, from invisible obstacles that need collision memory
L6	Small, qualitative evaluation: 1 simulated world, 10 real trials	The evaluation provides no TPR, false alarms/hour, latency or direction-error metrics, baselines or statistics

set of $\geq$50 h to verify false-alarm rates, and approximately 2,000 labeled collisions. Headless runs keep the cost of this data collection low, allowing overnight runs with 2–3 parallel containers.

III. CHALLENGES AND LIMITATIONS

These limitations raise two questions. Can we give MAV collision detection a guaranteed false-alarm rate in place of a hand-tuned threshold? Can the map retain invisible static obstacles while forgetting transient obstacles and false detections?

IV. PROPOSED METHOD AND CONTRIBUTIONS

We will reimplement Air Bumper as baseline B1 in ROS 2 with PX4 SITL. This port is not a contribution. It includes the per-axis detector (a^*, N), the sphere contact model $\mathbf{c}_c = \hat{\mathbf{C}}l$, the recovery setpoint $\mathbf{p}_r = (w\,\mathbf{G}/\|\mathbf{G}\| - (1-w)\hat{\mathbf{C}})R_d$ with $R_d = 2l$ sent directly as a PX4 offboard position setpoint, and permanent, lethal collision voxels in the map.

A. C1: Model-aware, calibrated collision detection (L1–L3)

(a) Residual signal. We compute

$$\mathbf{r}(t) = \mathbf{a}^{\mathrm{IMU}}(t) - \hat{\mathbf{a}}(t), \tag{1}$$

where $\hat{\mathbf{a}}$ is the acceleration explained by the MAV's commands. To obtain $\hat{\mathbf{a}}$, we pass the commanded velocity/acceleration setpoints through a closed-loop response model of the PX4 controller, identified from collision-free flights, then rotate the predicted acceleration into the body frame and remove gravity. This removes the contribution of braking and turning (L1, L3). Model-based residuals are a standard tool for collision detection on manipulators [11]; we apply them to IMU-based detection on MAVs.

(b) Features and score. Over a sliding window $W = 0.1$ s, we extract the peak 3D residual norm $\|\mathbf{r}\|$, which removes the per-axis issue in L2, along with the peak jerk, angular-rate residual and setpoint-tracking error. The tracking error captures pushing against an obstacle. A small, interpretable logistic regression model maps these features to a score $s \in [0, 1]$. We also test $s = \|\mathbf{r}\|$ as an ablation.

(c) Calibrated threshold. We apply split conformal calibration [12], [13] to n collision-free windows from the calibration worlds, setting

$$\hat{\tau} = s_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{2}$$

the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score. Under exchangeability, a new collision-free window triggers with probability $\leq \alpha$, giving a false-alarm rate set in advance of approximately $\alpha \cdot 3600/W$ per hour for non-overlapping windows. For 1 false alarm/h with $W = 0.1$ s, $\alpha \approx 2.8 \cdot 10^{-5}$ and $n \geq 1/\alpha \approx 36{,}000$ windows. This requires approximately 1 h of flight; we collect approximately 10 h. We target $\{1, 5, 20\}$ false alarms/h.

Because the windows are temporally correlated, we verify these rates empirically on the $\geq$50 h test set using exact Poisson (Clopper–Pearson-type) 95% intervals, with approximately 50 expected events at 1/h.

(d) Residual-based estimation. We compute the direction $\hat{\mathbf{C}}$ from $-\mathbf{r}$ in place of $-\mathbf{a}$, using azimuth and polar angle as in [1]. This removes the MAV's own acceleration from the direction estimate, recovery setpoint and mapped patch (L3).

B. C2: Evidence-aware collision memory with forgetting (L4, L5)

(a) LiDAR evidence. When the detector triggers, we check LiDAR points in a $\pm 10^\circ$ cone around the contact direction and assign one of three classes. A visible contact has a return within contact distance $+0.15$ m; the obstacle is already mapped and may be a person or moving object. An invisible contact has beams that pass beyond the contact point, as with glass. An unseen contact lies outside the vertical FOV, as with a stick hitting from below.

(b) Decaying belief. Each mark k stores its contact patch, hit count h_k and last-hit time t_k. Its belief decays as

$$b_k(t) = \exp\big(-(t - t_k)/\tau_{\text{class}}\big), \tag{3}$$

with $\tau_{\text{invisible}} = 600$ s, $\tau_{\text{unseen}} = 60$ s and $\tau_{\text{visible}} = 10$ s. The visible-contact memory lasts long enough to prevent an immediate repeat hit; LiDAR observations cover the obstacle after that. We tune all three values on calibration worlds and run a sensitivity study. A new hit within 0.3 m merges with the existing mark, increments the hit count (h_k+1) and resets t_k. Repeated contacts keep glass in memory, while memories of transient obstacles and false alarms fade.

(c) Graded costs and speed limits. The voxel cost is $c = c_{\max} b_k$, and a mark is lethal only if $h_k \geq 2$. An unconfirmed mark has a high cost but remains passable, so a single mark cannot make the goal unreachable. Near decayed or unconfirmed marks, we limit the setpoint speed to 0.3 m/s. Revisiting a marked location then causes at most a gentle cage contact that C1 detects quickly.

We will implement the method as Python (`rclpy`) nodes: `imu_residual`, `collision_detector`, `reaction`, `collision_memory` and `voxel_planner`. Memory only changes voxel costs and speed limits, so it can be added to any voxel-based planner.

We expect the following improvements over [1]: (1) a false-alarm rate set before deployment and verified on test worlds, with robustness to wind and aggressive maneuvers; (2) detection of diagonal and low-speed contacts; (3) smaller direction errors and more accurately placed recovery setpoints; (4) avoidance of phantom walls from transient obstacles and false alarms, at the cost of a small, measured increase in repeated collisions on static glass. RQ3 tests this trade-off.

V. Research questions and evaluation plan

RQ1 (Detection): Does C1 achieve a higher true-positive rate than Air Bumper's fixed per-axis threshold at the same false-alarm rate across speeds, maneuvers and wind? Does the calibrated rate hold under unseen gust profiles?

RQ2 (Estimation): How much does residual-based estimation reduce direction error and improve recovery distance?

RQ3 (Memory): Does C2 avoid phantom-wall failures with transient obstacles and false alarms, and how many repeat collisions on static glass does it add compared with permanent memory?

RQ4 (System): Does the full system increase mission success, reduce crashes and shorten time-to-goal compared with the baselines?

We will measure detection per collision event using ground truth from the cage contact sensor: TPR, false alarms per hour of collision-free flight (exact Poisson 95% intervals), empirical versus target false-alarm rate, detection latency and ROC-AUC. For estimation, we measure direction error (deg) and post-recovery distance to the obstacle (m).

Memory metrics are phantom volume (m^3 of marked cost with no real obstacle, integrated over time), repeated collisions within 0.5 m of an earlier collision, and blocked-goal events. System metrics are mission success, crashes (attitude $> 60^\circ$ or uncommanded altitude loss), collisions per run, time-to-goal and node run time ($<$5 ms per cycle).

The compute budget for E5 covers approximately 41 test

TABLE III
Baselines and ablations.

ID	Detection	Memory	Represents
B0	none (LiDAR-only navigation)	none	Standard MAV stack, which crashes as in [1] Fig. 4b
B1	Air Bumper [1]: $a^* = 20\,\text{m/s}^2$ and a^* re-tuned on calibration worlds	permanent, lethal	Primary paper
B2	Oracle (cage contact sensor)	permanent	Perfect detection to isolate the effect of memory
B3	C1	permanent	C1 alone
B4	Air Bumper	C2	C2 alone
Ours	C1	C2	Full method

TABLE IV
Planned experiments.

#	Experiment	Data	RQ
E1	Offline detection: ROC, TPR at matched false-alarm rate; features vs. $\|\mathbf{r}\|$ vs. raw acceleration; per speed and wind level	S1–S3 test logs	RQ1
E2	Calibration validity: target $\{1, 5, 20\}$/h vs. measured; in-distribution vs. unseen gusts/maneuvers	$\geq$50 h collision-free test set	RQ1
E3	Estimation: raw vs. residual direction; recovery distance	S1, S2	RQ2
E4	Memory (closed loop, oracle detector): permanent vs. decay-only vs. C2	S1, S2	RQ3
E5	Full comparison B0–B4 vs. Ours $\times$ 3 speeds $\times$ 3 seeds	S1–S3 test	RQ4
E6	Ablations/sensitivity: τ, lethal-after-h, speed limit, LiDAR evidence, W	Calibration worlds	RQ3, RQ4

worlds: $6 \times 3 \times 3 \times 41 \approx 2{,}200$ episodes, each lasting approximately 60 s, for approximately 37 h of simulated flight. At a PX4 SITL real-time factor of approximately 1, running 3 containers in parallel takes approximately 12 h of wall-clock time. If needed, we will reduce the full grid to one seed or two speeds while retaining all seeds for B1 versus Ours.

Worlds are the unit of statistical analysis. We will report 95% bootstrap confidence intervals over worlds, McNemar tests for success/failure and Wilcoxon signed-rank tests for continuous metrics, with Holm correction. All hyperparameters, including the B1 threshold, will be tuned only on calibration worlds.

Simulated contacts and IMU readings are idealized. We add noise/bias and report results for each noise level. Temporal correlation makes the conformal guarantee approximate, so we also report the empirical false-alarm rate. We do not test sim-to-real transfer.

During the first week, we will check whether collision-only models are invisible to the Gazebo `gpu_lidar` sensor and whether the caged x500 survives contacts in SITL without flipping. If the first check fails, we will use a LiDAR visibility mask. If the second fails, we will tune contact stiffness and lower the speed range.

VI. Team roles and individual contributions

TABLE V
Team roles.

Member	Role	Responsibilities and deliverables
Egemen Çevik	Simulation, scenarios and experiments	Docker/ROS 2/PX4 SITL/Gazebo setup; caged x500 with LiDAR, IMU, contact sensor; invisible-obstacle models; S1–S3 generators; voxel map and planner; episode runner, ground-truth labels; batch runs of E5
Cumali Sarı	Detection and estimation (C1)	Air Bumper detector for B1; controller-response model, residual node, features, logistic regression, conformal calibration, residual-based direction estimation; E1–E3; statistics
Ahmet Enes Çiğdem	Reaction and memory (C2)	Air Bumper recovery control and permanent memory for B1; collision-memory node, LiDAR evidence check, graded costs and speed limits; B3/B4 configs; E4, E6; figures

Each member will write and present their own module. We will document contributions through per-member git history and a weekly `CONTRIBUTIONS.md`.

In week 1 (11–16 Nov), we will have PX4 SITL and ROS 2 running and verify invisible obstacles and IMU collision spikes. Week 2 covers S1–S3, the voxel planner and B1; week 3 covers data collection and the controller-response model; week 4 covers C1 and E1–E3; week 5 covers C2 and E4; week 6 covers E5, E6 and statistics; week 7 is for the report and slides. The presentation is on 30.12.2026.

The minimum scope is B1 + C1 + E1–E3. Our target adds C2 and E4–E5. If time permits, we will also test a LiDAR model with partial glass returns to approximate real-world behavior and conduct a low-speed contact study.

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